

Solution of Exercise 6

Exercise 6.1 MPC stability

Consider the CLF inequality

$$\begin{aligned}\dot{V}(x) + l(x, -kx) &= px(-x^3 - kx) + qx^2 + rk^2x^2 \\ &= -px^4 - pkx^2 + qx^2 + rk^2x^2 \\ &= -px^4 - (pk - q - rk^2)x^2 \stackrel{!}{\leq} 0.\end{aligned}$$

The inequality is only fulfilled for all $x \in \mathbb{R}$, if

$$pk - q - rk^2 \stackrel{!}{\geq} 0.$$

For $k \rightarrow \pm\infty$ the equation is not satisfied, but might be in an interval with $pk > rk^2 + q$. Calculate the interval boundaries from $pk - q - rk^2 = 0$:

$$k_{1,2} = \frac{-p \pm \sqrt{p^2 - 4qr}}{-2r} = \frac{p \pm \sqrt{p^2 - 4qr}}{2r}.$$

It follows that $p \geq 2\sqrt{qr}$ must hold, otherwise there exists no $k \in \mathbb{R}$ for which the CLF inequality is satisfied for all $x \in \mathbb{R}$.

With additional input constraints $u \in [-1, 1]$, the following holds

$$|u| = |kx| = k|x| \leq 1 \quad \Rightarrow \quad |x| \leq \frac{1}{k}.$$

The smaller the gain k , the larger the domain of attraction. With

$$k = \frac{p - \sqrt{p^2 - 4qr}}{2r}.$$

follows the maximum domain of attraction

$$|x| \leq \frac{2r}{p - \sqrt{p^2 - 4qr}},$$

which can be enlarged by $p \uparrow$ or $r \uparrow$.

Exercise 6.2 Model predictive control with the toolbox GRAMPC

Procedure of the augmented Lagrangian method

- Optimization problem with scalar constraint $h(\mathbf{x}) \leq 0$
- Re-formulation as equality constraint $h(\mathbf{x}) + s = 0$ with $s \geq 0$
- Max-min-problem for augmented Lagrangian function

$$\begin{aligned} \max_{\lambda} \min_{\mathbf{u}, s \geq 0} \quad & L(\mathbf{u}, s, \lambda) = V(\mathbf{x}(T)) + \int_0^T l(\mathbf{x}, \mathbf{u}) + \lambda(h(\mathbf{x}) + s) + \frac{c}{2}(h(\mathbf{x}) + s)^2 dt \\ \text{s.t.} \quad & \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{x}(0) = \mathbf{x}_0 \\ & \mathbf{u} \in [\mathbf{u}^-, \mathbf{u}^+] \end{aligned}$$

- Minimization with regard to the slack variable s

$$s = \arg \min_{s \geq 0} \quad \lambda s + \frac{c}{2}(h(\mathbf{x}) + s)^2$$

can be solved analytically

$$0 = \lambda + c(h(\mathbf{x}) + s) \quad \Rightarrow \quad s = \max \left\{ 0, -h(\mathbf{x}) - \frac{\lambda}{c} \right\}$$

Equivalent equality constraints without slack variable

$$h^+(\mathbf{x}, \lambda) = \max \left\{ h(\mathbf{x}), -\frac{\lambda}{c} \right\} = 0$$

- Max-min-problem

$$\begin{aligned} \max_{\lambda} \min_{\mathbf{u}} \quad & L(\mathbf{u}, \lambda) = V(\mathbf{x}(T)) + \int_0^T l(\mathbf{x}, \mathbf{u}) + \lambda h^+(\mathbf{x}, \lambda) + \frac{c}{2}(h^+(\mathbf{x}, \lambda))^2 dt \\ \text{s.t.} \quad & \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{x}(0) = \mathbf{x}_0 \\ & \mathbf{u} \in [\mathbf{u}^-, \mathbf{u}^+] \end{aligned}$$

- Procedure of the augmented Lagrangian method for iterations $j = 1, \dots, j_{\max}$
 - a) Solve the minimization problem $(\mathbf{x}^{j+1}, \mathbf{u}^{j+1}) \leftarrow \min_{\mathbf{u}} L(\mathbf{u}, \lambda^j)$ s.t. system dynamics and constraints (e.g. with gradient method)
 - b) Solve the maximization problem via steepest ascent $\lambda^{j+1} = \lambda^j + c h^+(\mathbf{x}^{j+1}, \lambda^j)$
 - c) Optional adjustment of the penalty parameter c by heuristics
- Procedure of the gradient method for the iterations $i = 1, \dots, i_{\max}$
 - a) Forward integration of the system $\dot{\mathbf{x}}^i = \mathbf{f}(\mathbf{x}^i, \mathbf{u}^i)$ with initial conditions $\mathbf{x}^i(0) = \mathbf{x}_0$
 - b) Backward integration of the adjoint system $\dot{\boldsymbol{\lambda}}^i = -H_{\mathbf{x}}(\mathbf{x}^i, \mathbf{u}^i, \boldsymbol{\lambda}^i)$ with initial conditions $\boldsymbol{\lambda}^i(T) = V_{\mathbf{x}}(\mathbf{x}^i(T))$
 - c) Calculation of the gradient $\mathbf{g}^i = H_{\mathbf{u}}(\mathbf{x}^i, \mathbf{u}^i, \boldsymbol{\lambda}^i)$
 - d) Line search $\alpha^i = \arg \min_{\alpha} J(\Psi(\mathbf{u}^i - \alpha \mathbf{g}^i))$ with projection Ψ onto the admissible set $[\mathbf{u}^-, \mathbf{u}^+]$
 - e) Calculation of the new control input $\mathbf{u}^{i+1} = \Psi(\mathbf{u}^i - \alpha^i \mathbf{g}^i)$

Add-On 1 MPC for a two dimensional crane

a) State-space representation of the crane system

$$\left. \begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u_1 \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= u_2 \\ \dot{x}_5 &= x_6 \\ \dot{x}_6 &= -\frac{2x_4x_6 + u_1 \cos x_5 + g \sin x_5}{x_3} \end{aligned} \right\} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}). \quad (2)$$

b) By following the instructions in the provided MATLAB Live Script `crane_overhead.mlx`, the required analytical derivatives are obtained as follows:

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{2x_4x_6 + u_1 \cos x_5 + g \sin x_5}{x_3^2} & -\frac{2x_6}{x_3} & -\frac{g \cos x_5 - u_1 \sin x_5}{x_3} & -\frac{2x_4}{x_3} \end{bmatrix}, \quad (3)$$

$$\left(\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right)^T \mathbf{v}_f = \begin{bmatrix} 0 \\ v_{f,1} \\ \frac{2x_4x_6 + g \sin x_5 + u_1 \cos x_5}{x_3^2} v_{f,6} \\ v_{f,3} - \frac{2x_6}{x_3} v_{f,6} \\ \frac{u_1 \sin x_5 - g \cos x_5}{x_3} v_{f,6} \\ v_{f,5} - \frac{2x_4}{x_3} v_{f,6} \end{bmatrix}, \quad \mathbf{v}_f = [v_{f,1} \ v_{f,2} \ v_{f,3} \ v_{f,4} \ v_{f,5} \ v_{f,6}]^T$$

$$\frac{\partial \mathbf{f}}{\partial \mathbf{u}} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ -\frac{\cos x_5}{x_3} & 0 \end{bmatrix}, \quad \left(\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right)^T \mathbf{v}_f = \begin{bmatrix} v_{f,2} - \frac{\cos x_5}{x_3} v_{f,6} \\ v_{f,4} \end{bmatrix}. \quad (4)$$

For diagonal weighting matrices $\mathbf{Q} = \text{diag}(q_1, \dots, q_6)$ and $\mathbf{R} = \text{diag}(r_1, r_2)$, the gradients of the integral cost are given by

$$\frac{\partial I}{\partial \mathbf{x}} = \begin{bmatrix} q_1(x_1 - x_{1,\text{des}}) \\ q_2(x_2 - x_{2,\text{des}}) \\ q_3(x_3 - x_{3,\text{des}}) \\ q_4(x_4 - x_{4,\text{des}}) \\ q_5(x_5 - x_{5,\text{des}}) \\ q_6(x_6 - x_{6,\text{des}}) \end{bmatrix}, \quad \frac{\partial I}{\partial \mathbf{u}} = \begin{bmatrix} r_1(u_1 - u_{1,\text{des}}) \\ r_2(u_2 - u_{2,\text{des}}) \end{bmatrix}. \quad (5)$$

Considering the parabola constraint on load of crane and limit on angular velocity, the constraints vector and respectively the gradients are given by

$$\mathbf{h}(\mathbf{x}) = \begin{bmatrix} x_3 \cos x_5 - a(x_1 + x_3 \sin x_5)^2 - b \\ x_6 - \dot{\phi}_{\max} \\ -\dot{\phi}_{\min} - x_6 \end{bmatrix}, \quad (6)$$

$$\frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \begin{bmatrix} -2as_1 & 0 & \cos x_5 - 2a s_1 \sin x_5 & 0 & -x_3 \sin x_5 - 2a s_1 x_3 \cos x_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix},$$

$$\left(\frac{\partial \mathbf{h}}{\partial \mathbf{x}}\right)^T \mathbf{v}_h = \begin{bmatrix} -2as_1 v_{h,1} \\ 0 \\ (\cos x_5 - 2a s_1 \sin x_5) v_{h,1} \\ 0 \\ (-x_3 \sin x_5 - 2a s_1 x_3 \cos x_5) v_{h,1} \\ v_{h,2} - v_{h,3} \end{bmatrix}, \quad (7)$$

$$\frac{\partial \mathbf{h}}{\partial \mathbf{u}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \left(\frac{\partial \mathbf{h}}{\partial \mathbf{u}}\right)^T \mathbf{v}_h = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (8)$$

c) The crane can successfully reach the desired state while avoiding the obstacle when:

- the prediction horizon is sufficiently long,
- the number of horizon discretization points is sufficiently high,
- the angle and angular-velocity limits are not excessively restrictive,
- the obstacle geometry yields a feasible path,
- the input limits allow the required trolley and hoisting motion,
- and the optimization can be completed within the available sampling time.

Figure 1 compares two runs of the second scenario with $\phi_{\max} = 10^\circ$ and $\phi_{\max} = 1^\circ$. The stricter angle limit reduces the permitted load swing and therefore causes the trolley to move more slowly toward the desired position.

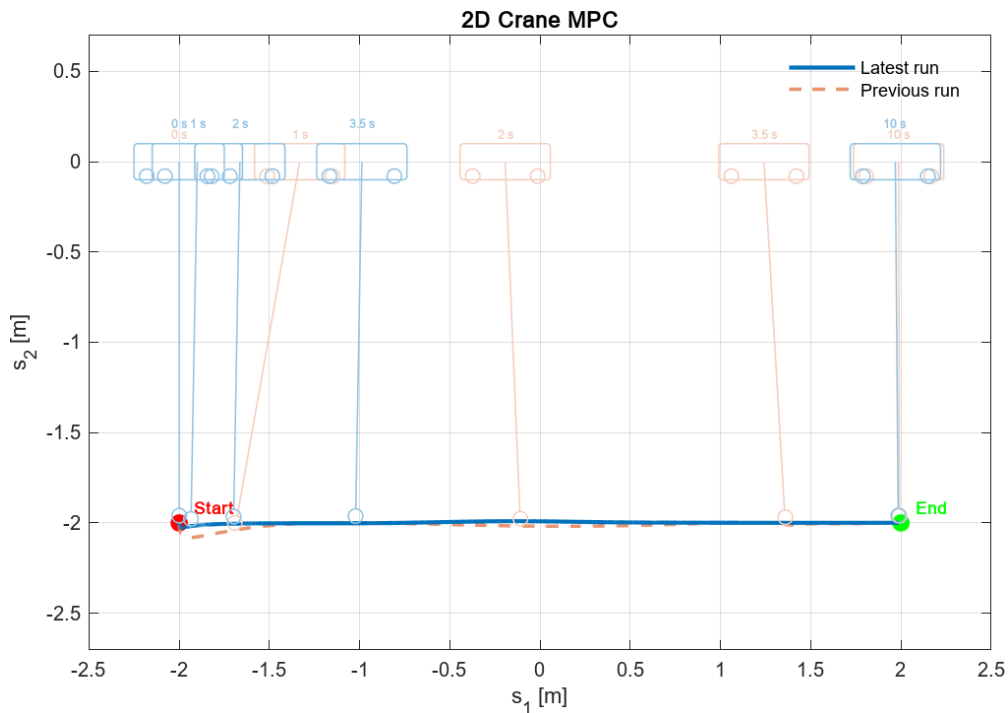


Figure 1: Comparison of two MPC simulations using different parameter settings.

Add-On 2 MPC for an evasive maneuver

a) Optimization problem for the evasive maneuver

$$\begin{aligned} \min_{\mathbf{u}} \quad & J(\mathbf{u}) = \frac{1}{2} \Delta \mathbf{x}(T)^T \mathbf{S} \Delta \mathbf{x}(T) + \int_0^T \frac{1}{2} \Delta \mathbf{x}^T \mathbf{Q} \Delta \mathbf{x} + \frac{1}{2} \mathbf{u}^T \mathbf{R} \mathbf{u} d\tau \\ \text{s.t.} \quad & \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{x}(0) = \mathbf{x}_0 \\ & \mathbf{u} \in [\mathbf{u}^-, \mathbf{u}^+] \\ & (x_1 - x_{p,1})^2 + (x_2 - x_{p,2})^2 \geq (3 \text{ m})^2 \end{aligned}$$

with $\Delta \mathbf{x} = \mathbf{x} - \mathbf{x}_f$, time horizon $T = 1 \text{ s}$ and weighting matrices

$$\begin{aligned} \mathbf{S} &= \mathbf{Q} = \text{diag}([1 \quad 1 \quad 1]) \\ \mathbf{R} &= \text{diag}([0.01 \quad 0.01]) \end{aligned}$$

b) Gradient of the system dynamics and multiplications with $\mathbf{v} = [v_1, v_2, v_3]^T \in \mathbb{R}^{n_x}$:

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} 0 & 0 & -u_1 \sin x_3 \\ 0 & 0 & u_1 \cos x_3 \\ 0 & 0 & 0 \end{bmatrix}, \quad \left(\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right)^T \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ -u_1 \sin x_3 v_1 + u_1 \cos x_3 v_2 \end{bmatrix} \quad (11)$$

$$\frac{\partial \mathbf{f}}{\partial \mathbf{u}} = \begin{bmatrix} \cos x_3 & 0 \\ \sin x_3 & 0 \\ 0 & 1 \end{bmatrix}, \quad \left(\frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right)^T \mathbf{v} = \begin{bmatrix} \cos x_3 v_1 + \sin x_3 v_2 \\ v_3 \end{bmatrix} \quad (12)$$

Gradients of the integral term $I(\mathbf{x}, \mathbf{u}) = \frac{1}{2} (Q_1 \Delta x_1^2 + Q_2 \Delta x_2^2 + Q_3 \Delta x_3^2 + R_1 u_1^2 + R_2 u_2^2)$, where $\mathbf{Q} = \text{diag}(Q_1, Q_2, Q_3)$ and $\mathbf{R} = \text{diag}(R_1, R_2)$:

$$\frac{\partial I}{\partial \mathbf{x}} = \begin{bmatrix} Q_1 \Delta x_1 \\ Q_2 \Delta x_2 \\ Q_3 \Delta x_3 \end{bmatrix}, \quad \frac{\partial I}{\partial \mathbf{u}} = \begin{bmatrix} R_1 u_1 \\ R_2 u_2 \end{bmatrix} \quad (13)$$

Gradient of the terminal cost $V(\mathbf{x}) = \frac{1}{2} (S_1 \Delta x_1^2 + S_2 \Delta x_2^2 + S_3 \Delta x_3^2)$, where $\mathbf{S} = \text{diag}(S_1, S_2, S_3)$:

$$\frac{\partial V}{\partial \mathbf{x}} = \begin{bmatrix} S_1 \Delta x_1 \\ S_2 \Delta x_2 \\ S_3 \Delta x_3 \end{bmatrix} \quad (14)$$

Gradient of the constraint $h(\mathbf{x}) = 3^2 - (x_1 - x_{p,1})^2 - (x_2 - x_{p,2})^2 \leq 0$ and multiplications with $\mathbf{v} = [v_1]^T \in \mathbb{R}^{n_h}$:

$$\frac{\partial h}{\partial \mathbf{x}} = [-2(x_1 - x_{p,1}) \quad -2(x_2 - x_{p,2}) \quad 0], \quad \left(\frac{\partial h}{\partial \mathbf{x}} \right)^T \mathbf{v} = \begin{bmatrix} -2(x_1 - x_{p,1})v_1 \\ -2(x_2 - x_{p,2})v_1 \\ 0 \end{bmatrix} \quad (15)$$

$$\frac{\partial h}{\partial \mathbf{u}} = [0 \quad 0], \quad \left(\frac{\partial h}{\partial \mathbf{u}} \right)^T \mathbf{v} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (16)$$

c) Under which conditions can the evasion fail?

- MPC runs into local minimum, which does not correspond to any admissible solution.
- Constraint is violated because optimization problem is solved only suboptimally.
- Time horizon is too short to avoid collision.